

Agile Software Engineering and DevOps – Lab 2

Title - Applying Agile Scrum Methodology to an Existing Waterfall Project

Part A – Role Assignment & Project Reframing

Task A1: Assign Scrum Roles

ROLES	NAMES	RESPONSIBILITIES
Product Owner	Shradha S Shetty	Defines and prioritizes product backlog.
Scrum Master	Shristi Nagesh T	Ensures Scrum practices are followed and removes impediments for the team.
Development Team	i. Shaikh Abrar Ahmed ii. Samya iii. Soubhagya S J	Design, develop and test product and identify potential product increment in each sprint.

Task A2: Reframe the Project for Agile

1. What was assumed as fixed in Waterfall?

The system features such as user authentication, attendance management, assignments, CIA tracking, analytics, notifications, and user roles were assumed to be fully defined. The overall system structure, modules, and workflows were treated as stable and unchanging throughout development.

2. What can now be incremental?

Features like analytics dashboards, notifications, UI design, performance optimization, security enhancements, resource sharing, and system usability improvements can be developed and improved in small increments over multiple iterations based on user feedback.

3. Why Agile suits this project?

Agile suits this project because the system requires continuous improvement and regular feature enhancements. It allows incremental development of modules such as analytics, notifications, and UI instead of building everything at once. Agile supports early testing, faster feedback from users, and flexible adaptation to changing requirements. This makes the system more scalable, user-centered and involved, and easier to improve over time.

Part B – Product Backlog Creation

Task B1: Write User Stories

Waterfall Requirements:

- User login and signup
- Role-based access: Admin / Faculty / Student
- Attendance marking
- Analytics dashboard for attendance and tasks
- Notifications and alerts:
- Resource and material sharing
- Security through authentication and authorization

Task B1: User Stories

1. As a student, I want to register an account so that I can access the system.
2. As a user, I want to log in securely so that I can use my personal dashboard.
3. As a student, I want to check my attendance daily so that my presence is recorded.
4. As a student, I want to update my daily progress so that my academic work is tracked.
5. As a faculty member, I want to view and mark student attendance so that I can monitor participation.
6. As a faculty member, I want to create assignments so that students can complete and submit them.
7. As a faculty member, I want to evaluate assignments so that student performance can be assessed.
8. As a faculty member, I want to record CIA and mock marks so that academic records are maintained.
9. As a faculty member, I want to view analytics dashboards so that I can analyze student progress.
10. As a student, I want to view my performance analytics so that I can understand my progress.
11. As a user, I want to access department contact details so that I can communicate with the correct department.
12. As a user, I want to receive notifications so that I do not miss important academic updates.

Task B2: Prioritize the Product Backlog

PRIORITY	USER STORY
High	As a student, I want to mark my attendance daily so that my presence is recorded.
High	As a faculty member, I want to create assignments so that students can complete and submit them.
High	As a faculty member, I want to record CIA and mock marks so that academic records are maintained.
High	As a user, I want to access department contact details so that I can communicate with the correct department.
High	As a faculty member, I want to view analytics dashboards so that I can analyze student progress.
Medium	As a user, I want to receive notifications so that I do not miss important updates.
Medium	As a student, I want to update my daily progress so that my academic work is tracked.
Low	As a user, I want to customize my profile so that my personal information is updated.
Low	As a user, I want to view system announcements so that I stay informed about platform updates.

Part C – Sprint Planning

Task C1: Sprint Definition

- Sprint Duration: 2 Weeks
- Sprint Goal:
To actively develop and implement the core modules of the system by focusing on building, and integrating the main functionalities including attendance management, assignment handling, CIA evaluation, departmental support access, and analytics dashboards.

Task C2: Sprint Backlog Creation

User Story 1: As a student, I want to register an account so that I can access the system.

Tasks:

- Create user table in database
- Implement backend registration API
- Validate inputs and store data

Owner: Backend Developer

User Story 2: As a user, I want to log in securely so that I can use my personal dashboard.

Tasks:

- Implement authentication API
- Password validation
- Role-based redirection

Owner: Backend Developer

User Story 3: As a student, I want to check my attendance daily so that my presence is recorded.

Tasks:

- Create attendance table
- Build attendance checking API

Owner: Backend Developer

User Story 4: As a student, I want to update my daily progress so that my academic work is tracked.

Tasks:

- Create daily progress table
- Implement progress submission API

- Display progress history

Owner: Backend Developer

User Story 5: As a faculty member, I want to create assignments so that students can complete and submit them.

Tasks:

- Create assignments table
- Build assignment creation API
- Build assignment submission flow

Owner: Backend Developer

TEAM MEMEBRS:

1. Samya - 1RUA24CSE0405
2. Shaikh Abrar Ahmed - 1RUA24CSE0418
3. Shradha S Shetty - 1RUA24CSE0435
4. Shristi Nagesh T - 1RUA24CSE0449
5. Soubhagya S J - 1RUA24CSE0469